Algebra SEP Worksheet 7

8/8/23

Rings and Modules

Setup

1. Find an example of each of the following.

(a) A ring R and a left ideal $I \subset R$ that is *not* a right ideal.

(b) A ring R and a prime ideal $P \subset R$ that is not maximal. (Let's make this harder and say $P \neq \{0\}$.)

(c) A ring R, an ideal I, and a prime ideal $P \subset R/I$.

(d) A ring R and an ideal P that is primary, but not prime.

(e) A map of rings $f\colon R \to S$ and a prime ideal $P \subset R$ such that $f(P)S \subset S$ is *not* prime. (Can you find an example where $f(P)S \neq (1)$?)

2. (January 2020 Problem 3(a) (modified)) Let R be a commutative ring, I an ideal in R, and S a multiplicative subset of R. There are two standard correspondences involving prime ideals:

(i) Prime ideals in R/I are in bijection with prime ideals in R which contain I.

(ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S.

Prove (i) and (ii).

3. Let R be a local ring with maximal ideal $\mathfrak{m}$. Show that $x \in R$ is a unit if and only if $x \notin \mathfrak{m}$.

4. Let R be a commutative ring, and S a multiplicative subset of R.

(a) When is $S^{-1}R = 0$?

(b) When is the localization map $R \to S^{-1}R$ injective?

Quick notational aside: remember that for a commutative ring R and an element $f \in R$, we write R_f for the localization of R at the multiplicative set $\{f^n : n \geq 0\}$. HOWEVER! This is confusing because for a prime ideal $P \subset R$, we write R_P for the localization of R at the multiplicative set $R \setminus P$. So, $\mathbb{C}[x]_x$ is different from $\mathbb{C}[x]_{(x)}$. Worse yet, people often write $\mathbb{Z}_p$ to mean either the p-adic integers or $\mathbb{Z}/p\mathbb{Z}$.

Punchline

5. (August 2019 Problem 2) This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I, denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

(a) Prove that if I is primary, then $\sqrt{I}$ is prime.

(b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x, y]$ primary? Be sure to carefully justify your answer.

6. (January 2018 Problem 2, modified) Consider the ring $R = \mathbb{C}[x]/(x^3)$.

(a) Give an example of a nonzero zero-divisor in R.

(b) Show that the set of zero-divisors in R form an ideal.

(c) Let S be a commutative ring. Prove that if the set of zero-divisors is an ideal I, then $I \subset S$ is a prime ideal.

7. (January 2018 Problem 1)

(a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.

(b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \to R_f$ is neither injective nor surjective.

(c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

8. (January 2020 Problem 3(b)-(c))

(a) Give an example of a ring with finitely many elements that has exactly three prime ideals.

(b) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

9. Consider $M_2(\mathbb{Z})$, the ring of 2×2 matrices with entries in $\mathbb{Z}$.

(a) **(Aug 2019 1(b) and Jan 2021 1(b))** Find a 2-sided ideal of $M_2(\mathbb{Z})$ that is neither 0 nor the whole ring.

(b) **(Jan 2018 2(b), modified)** Give an example of a nonzero left zero-divisor of $M_2(\mathbb{Z})$.

(c) **(Jan 2018 2(c), modified)** Show that the left zero-divisors of $M_2(\mathbb{Z})$ do not form a left ideal.

(d) Find all the prime ideals of $M_2(\mathbb{Z})$. (For a noncommutative ring, a proper 2-sided ideal P is prime if whenever A, B are 2-sided ideals such that $AB \subset P$, then either $A \subset P$ or $B \subset P$.)